MAT 534 HW05

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Problem 1. Let $\mathscr{C}$ be a category.

(a) Show that for any object $X \in \mathrm{Ob}(\mathscr{C})$, there is only one morphism from X to itself that has the defining property of id_X.

(b) Suppose that $f \in \mathrm{Hom}(X, Y)$ is an isomorphism. Does this imply that f has a unique inverse in $\mathrm{Hom}(Y, X)$?

(c) Show that the group $\mathrm{Aut}(X)$ acts on the set of isomorphisms in $\mathrm{Hom}(X, Y)$ and that this action is transitive.

Solution. (a) Let id_X and id'_X be two such morphisms. Then

$$\mathrm{id}_X = \mathrm{id}_X \circ \mathrm{id}'_X = \mathrm{id}'_X .$$

(b) Yes; if $g, g' \in \mathrm{Hom}(Y, X)$ are two inverses of f, meaning $g \circ f = g' \circ f = \mathrm{id}_X$ and $f \circ g = f \circ g' = \mathrm{id}_Y$, then

$$g = g \circ \underbrace{(f \circ g')}_{\mathrm{id}_Y} = \underbrace{(g \circ f)}_{\mathrm{id}_X} \circ g' = g'.$$

(c) Define an action of $\mathrm{Aut}(X)$ on the isomorphisms in $\mathrm{Hom}(X, Y)$ by $f \cdot g = f \circ g$. This is a right action because

$$f \cdot (g \circ h) = f \circ (g \circ h) = (f \circ g) \circ h = (f \cdot g) \cdot h, \quad f \cdot \mathrm{id}_X = f \circ \mathrm{id}_X = f.$$

The action is transitive because given $f_1, f_2 \in \mathrm{Hom}(X, Y)$ are isomorphisms then $f_2^{-1} \circ f_1 \in \mathrm{Aut}(X)$ satisfies

$$f_2 \cdot (f_2^{-1} \circ f_1) = f_2 \circ f_2^{-1} \circ f_1 = f_1.$$

□

Problem 2. Given morphism $f \colon X \to Y$, denote by $(*)$ the property that if $g \colon Y \to Z$ and $h \colon Y \to Z$ are two morphisms such that $g \circ f = h \circ f$, then $g = h$.

(a) Show that in ((sets)), $f \colon X \to Y$ is surjective iff f satisfies $(*)$.

(b) Characterize the morphisms satisfying $(*)$ in ((topological spaces)).

(c) Characterize the morphisms satisfying $(*)$ in ((groups)).

(d) Characterize the morphisms satisfying $(*)$ in ((abelian groups)).

Solution. (a) If f is surjective, then for each $y \in Y$, there exists $x \in X$ such that $y = f(x)$, so $g(y) = g(f(x)) = h(f(x)) = h(y)$ for each $y \in Y$, which means $g = h$.

On the other hand if f isn't surjective, then there exists $y_0 \in Y$ that is not mapped to by any $x \in X$. If $g = \mathrm{id}_Y$ and h is the function mapping y_0 to any other point in Y while fixing all the other points (i.e. $h(y_0) = y_1$ for some $y_1 \neq y_0$ and $h|_{Y \setminus \{y_0\}} = \mathrm{id}_{Y \setminus \{y_0\}}$). Then, $g \neq h$ but $g \circ f = h \circ f$ because y_0 is never passed as a value to either g or h. Thus if f isn't surjective then $(*)$ doesn't hold.

todo

(b)

□

Problem 3. Do the following categories have an inital or terminal object? If so, describe them; if not, explain why not.

(a) $(\!(\text{sets})\!)$

(b) $(\!(\text{groups})\!)$

(c) $(\!(\text{fields})\!)$

(d) $(\!(\text{fields of characteristic } 0)\!)$

Solution. (a) $(\!(\text{sets})\!)$ has initial object $\varnothing$, since there is tautologically one function $\varnothing \to A$ for any set A (and no other set is an initial object because a morphism $X \to \{1,2\}$ necessarily has at least two possible functions by sending everything to either 1 or 2). Any singleton set is a final object because a function from X to a singleton set leaves exactly one choice for each element of X (and no other set is a final object because a morphism $f\colon \{1\} \to A$ either has no choices for $f(1)$ if A is empty, or at least two choices for $f(1)$ if A has more than one element).

(b) Any group with one element is both an initial and final object, since a morphism of groups always maps the identity of one to the identity of the other. Since all initial objects are isomorphic and all final objects are isomorphic (by Problem 6), this characterizes all initial and final objects in $(\!(\text{groups})\!)$.

(c) The category of fields does not have an initial object or a final object: any morphism of fields is injective, so there is no morphism between fields of different characteristics, i.e. given any field F there exists another field F' (for example any field with a different characteristic) for which $\mathrm{Hom}(F, F')$ is empty, which automatically means $\mathrm{Hom}(F, G)$ cannot have exactly one element for all fields G.

(d) The category of fields with characteristic zero has (any field isomorphic to) $\mathbb{Q}$ as an initial object, since a morphism $\mathbb{Q} \to F$ is uniquely determined due to the fact that 1 must map to 1_F. On the other hand, there is no final object, because apparently you can construct fields of arbitrarily large cardinality (I don't understand why well enough to write down why), and if F, F' are fields such that F' has larger cardinality than F then there is no morphism $F' \to F$ because field homomorphisms must be injective, so F cannot be final. □

Problem 4. In some categories, the objects are sets and have elements, and then we can try to describe those elements category-theoretically. For example:

(a) Let $\mathbf{1}$ be any one-element set in $(\!(\text{sets})\!)$, for example $\{\varnothing\}$. Show that the elements of a set X are in 1-to-1 correspondence with the morphisms $\mathrm{Hom}(\mathbf{1}, X)$.

(b) In $(\!(\text{groups})\!)$, show that the elements of a group G are in 1-to-1 correspondence with $\mathrm{Hom}(\mathbb{Z}, G)$.

(c) Let F be a field, and consider the category $(\!(F\text{-vector spaces})\!)$ of vector spaces over the field F. Can you find a vector space V_0 in this category such that for any vector space V over F, the elements of V are in 1-to-1 correspondence with $\mathrm{Hom}(V_0, V)$?

Solution. (a) Since $\mathbf{1} = \{\varnothing\}$ has only one element, a morphism $f\colon \mathbf{1} \to X$ is uniquely determined by $f(\varnothing)$. Thus there is a correspondence between elements $x \in X$ and morphisms $f \in \mathrm{Hom}(\mathbf{1}, X)$ given by

$$x \longleftrightarrow (\varnothing \mapsto x).$$

(b) Since $\mathbb{Z}$ is generated by 1, a morphism $f\colon \mathbb{Z} \to G$ is uniquely determined by $f(1)$, as this tells us $\phi(n) = g^n$ for all $n \in \mathbb{Z}$. Thus there is a correspondence between elements $g \in G$ and morphisms $\phi \in \mathrm{Hom}(\mathbb{Z}, G)$ given by

$$g \longleftrightarrow (n \mapsto g^n).$$

(c) Let $V_0 = F$. Since V_0 is a 1-dimensional F-vector space, a morphism $f \in \mathrm{Hom}(V_0, V)$ is determined uniquely by $f(1)$. Thus there is a correspondence between elements $v \in V$ and morphisms $f \in \mathrm{Hom}(V_0, V)$ given by

$$v \longleftrightarrow (c \mapsto cv).$$

□

Problem 5. Let $\mathscr{C}$ be a category, and $A \in \mathscr{C}$ a fixed object. For every $X \in \mathrm{Ob}(\mathscr{C})$, let $F(X) = \mathrm{Hom}(A, X)$ be the set of morphisms from A to X in our category. Show that F is a functor from $\mathscr{C}$ to ((sets)).

Solution. If $f\colon X \to Y$ is a morphism in $\mathscr{C}$, let $F(f)$ be the morphism $F(X) \to F(Y)$ sending

$$(g\colon A \to X) \mapsto (f \circ g\colon A \to Y)$$

where $g \in \mathrm{Hom}(A, X)$ is any morphism. This is a functor because

- $F(\mathrm{id}_X)$ is the map $(g\colon A \to X) \mapsto (\mathrm{id}_X \circ g = g\colon A \to X)$ which is $\mathrm{id}_{F(X)}$.
- If $f_1 \in \mathrm{Hom}(X, Y)$ and $f_2 \in \mathrm{Hom}(Y, Z)$ then $F(f_2) \circ F(f_1)$ is the composed map

  $$(g\colon A \to X) \mapsto (f_1 \circ g\colon A \to Y) \mapsto (f_2 \circ f_1 \circ g\colon A \to Z)$$

  where $g \in \mathrm{Hom}(A, X)$, whose net effect is exactly that of $F(f_2 \circ f_1)\colon F(X) \to F(Z)$, i.e. F is well-behaved under composition. □

Problem 6. Let A and B be two initial (or final) objects in a category. Show that there is a unique isomorphism between A and B.

Solution. If A and B are two initial objects or two final objects, then the important data is that $\mathrm{Hom}(A, A)$, $\mathrm{Hom}(A, B)$, $\mathrm{Hom}(B, A)$, and $\mathrm{Hom}(B, B)$ each have exactly one morphism, which we will respectively call $f_{AA} = \mathrm{id}_A$, f_{AB}, f_{BA}, and $f_{BB} = \mathrm{id}_B$. Then

$$\begin{aligned} f_{AB} \circ f_{BA} \in \mathrm{Hom}(B, B) &\implies f_{AB} \circ f_{BA} = \mathrm{id}_B, \\ f_{BA} \circ f_{AB} \in \mathrm{Hom}(A, A) &\implies f_{BA} \circ f_{AB} = \mathrm{id}_A, \end{aligned}$$

so f_{AB} is an isomorphism, and in fact it is the unique (iso)morphism from A to B. □

Problem 7. Given a topological space X, we can define a category $\mathscr{C}_X$ as follows: The objects of $\mathscr{C}_X$ are the open subsets of X, and for any two open subsets $U, V \subseteq X$, the set of morphisms in $\mathscr{C}_X$ is

$$\mathrm{Hom}(U, V) = \begin{cases} \{i_{UV}\} & \text{if } U \subseteq V, \\ \varnothing & \text{if } U \not\subseteq V. \end{cases}$$

If $U \subseteq V \subseteq W$, we define composition as $i_{VW} \circ i_{UV} = i_{UW}$.

(a) Prove that $\mathscr{C}_X$ is indeed a category.

(b) Does $\mathscr{C}_X$ have an initial or final object?

(c) For every open $U \subseteq X$, define $F(U)$ to be the set of continuous functions from U to $\mathbb{R}$. Show that F is a contravariant functor from $\mathscr{C}_X$ to the category (($\mathbb{R}$-vector spaces)).

Solution. (a) To show that $\mathscr{C}$ is a category, we only need to check that the morphisms behave as they are supposed to. Each object has identity i_{UU} (because inclusion of U into U is the identity), and if $U \subseteq V \subseteq W \subseteq Z$ are open subsets then

$$(i_{WZ} \circ i_{VW}) \circ i_{UV} = i_{VZ} \circ i_{UV} = i_{UZ} = i_{WZ} \circ i_{UW} = i_{WZ} \circ (i_{VW} \circ i_{UV}),$$

proving associativity.

(b) Each $\mathrm{Hom}(U, V)$ has at most one morphism, so the only way that U is initial is if $U \subseteq V$ for all open V, i.e. $U = \varnothing$, so $\boxed{\varnothing}$ is the unique initial object; and the only way that V is final is if $V \supseteq U$ for all open U, i.e. $V = \boxed{X}$ is the unique final object.

(c) If $U, V \subseteq X$ are open sets such that $U \subseteq V$, then the map $F(V) \to F(U)$ given by

$$\begin{aligned} F(i_{UV}) : F(V) &\to F(U) \\ (f \colon V \to \mathbb{R}) &\mapsto (f \circ i_{UV} \colon U \to \mathbb{R}) \end{aligned}$$

is well-defined because inclusion is a continuous map and is obviously linear since $(f + g) \circ i_{UV} = f \circ i_{UV} + g \circ i_{UV}$ and $(cf) \circ i_{UV} = c(f \circ i_{UV})$ essentially by definition, so we will say F sends i_{UV} to this morphism between $\mathbb{R}$-vector spaces.

Now, we check that F is a contravariant functor:

- For any open $U \subseteq X$, $F(\mathrm{id}_U) = F(i_{UU})$ is the map $(f \colon U \to \mathbb{R}) \mapsto (f \circ i_{UU} = f \colon U \to \mathbb{R})$, so it is the identity on $F(U)$.
- If $U \subseteq V \subseteq W$ are open sets in X, then $F(i_{UV}) \circ F(i_{VW})$ is the composed map

  $$(f \colon W \to \mathbb{R}) \mapsto (f \circ i_{VW} \colon V \to \mathbb{R}) \mapsto (f \circ i_{VW} \circ i_{UV} \colon U \to \mathbb{R})$$

  which is the same as the map $F(i_{VW} \circ i_{UV})$ by definition.

$\square$